

The Tom Klootwijk Manifold

A rigorous seven-dimensional "round-factor" manifold

Product: $\mathcal{K}_{\text{TK}}(\mathbf{r}) := \prod_{i=0}^6 S^1(r_i), \quad r_i > 0.$

Implicit: $\mathcal{K}_{\text{TK}}(\mathbf{r}) = \{x \in \mathbb{R}^{14} : x_{2i}^2 + x_{2i+1}^2 = r_i^2, i = 0, \dots, 6\}.$

Parametric: $\Phi_\alpha(\theta)_{2i:2i+1} = r_i(\cos(\theta_i + \phi_i), \sin(\theta_i + \phi_i)).$

Metric/frame: $g = \sum_{i=0}^6 r_i^2 d\theta_i^2, \quad E_i = r_i^{-1} \partial_{\theta_i}, \quad g(E_i, E_j) = \delta_{ij}.$

Formal type

Proof kernel

Seven-way frame

Unity literal

Corrections

A smooth, compact, connected 7-manifold, diffeomorphic to the 7-torus T^7 . It is intrinsically flat under the product metric and extrinsically curved in $\mathbb{R}^{14}$.
With $F_i = x_{2i}^2 + x_{2i+1}^2 - r_i^2$, the seven rows of DF have disjoint nonzero supports on $F^{-1}(0)$, so $\text{rank } DF = 7$. The regular-value theorem gives dimension $14 - 7 = 7$.
The global unit tangent fields $E_0, \dots, E_6$ satisfy $g(E_i, E_j) = \delta_{ij}$ at every point. This is the precise meaning of seven mutually orthogonal directions.
 $X(u, v, t) = P\Phi_\alpha((m_i u + n_i v + \omega_i t)_{i=0}^6)$ renders a 3D linear projection of a periodic 2D slice. The projection is a visualization, not the whole 7D object.
Indexing cannot change Hamming distance: $943 \oplus 937 = 6 = 110_2$, hence distance 2. Also, "SDF = 0" denotes a level set; it does not by itself create a sphere or constant-time renderer.

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